Task 1 (Interpolation) **1 + 3 + 1 = 5 points**

Consider the function

$$f(x) = \frac{x+1}{x^2+1}$$

and the grid points $x_0 = -1$, $x_1 = 0$, $x_2 = 1$.

(a) Write down the Vandermonde matrix for these grid points. Decide if the resulting matrix is invertible and explain your answer.

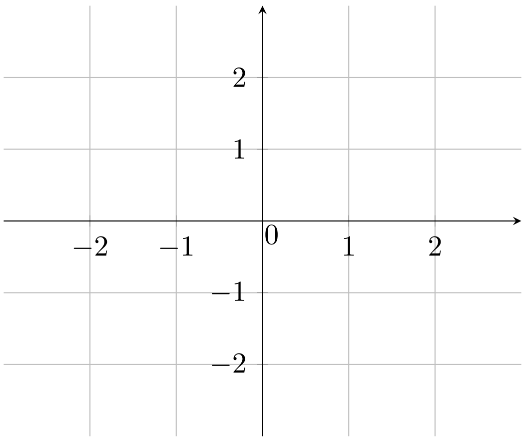
(b) Compute the interpolation polynomial for the data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $(x_2, f(x_2))$ and write it in the form

$$p(x) = \alpha_N x^N + \alpha_{N-1} x^{N-1} + \cdots \alpha_1 x + \alpha_0.$$

Sketch the data points and the interpolation polynomial in Figure 1.

(b) Assume now that you only use the data points $(x_1, f(x_1))$ and $(x_2, f(x_2))$. Discuss the expected, the maximal and the actual degree of the interpolation polynomial.

Figure 1:



Task 2 (Numerical integration) **3 + 2 = 5 points**

a) Consider the integral $I(f) = \int_a^b f(x)dx$ for a general function $f : [a,b] \to \mathbb{R}$. Derive the quadrature formula for the trapezoid rule in order to approximate the integral $I(f)$. For this, explain shortly the idea using a sketch and make use of an appropriate interpolation.

b) Consider now the function $f(x) = c_1x^4 + c_2x^3 + c_3x + c_4$.

   - Estimate the quadrature error for the trapezoid rule on the interval $[-1,3]$ by a smallest possible upper bound for $c_1 = -\frac{1}{12}, c_2 = \frac{1}{3}, c_3 = 7, c_4 = -2$.
   - Which quadrature error do you expect for $c_1 = c_2 = 0, c_3, c_4 \in \mathbb{R}$? Explain your answer.

Task 3 (Romberg extrapolation, numerical differentiation) **3 + 2 = 5 points**

a) Consider the forward difference quotient to approximate the first-order derivative of a function $f : \mathbb{R} \to \mathbb{R}$. Perform one step of the Romberg scheme and state the resulting calculation formula. What is the convergence order of the resulting scheme? Explain your answer.

b) Approximate the first-order derivative of $f(x) = x^{-1}$ at $x = 1$ by the forward difference quotient, the resulting scheme from (a) and the central difference quotient for $h = \frac{1}{2}$. Compare the approximations with the actual value.

Task 4 (Nonlinear equations) **1 + 2 + 2 = 5 points**

a) Derive the Newton's method for finding zeros of a sufficiently smooth, general function f starting from the Taylor expansion.

b) Calculate the first two steps of Newton's method for finding a zero of the function $f(x) = \frac{1}{4}x^2$ starting from $x_0 = 4$. Explain the iteration scheme shortly with words and using a sketch in Figure 2.

c) Explain the connection between Newton's method and Banach's fixed point iteration. Determine the convergence order of Newton's method and explain your result.

Figure 2: